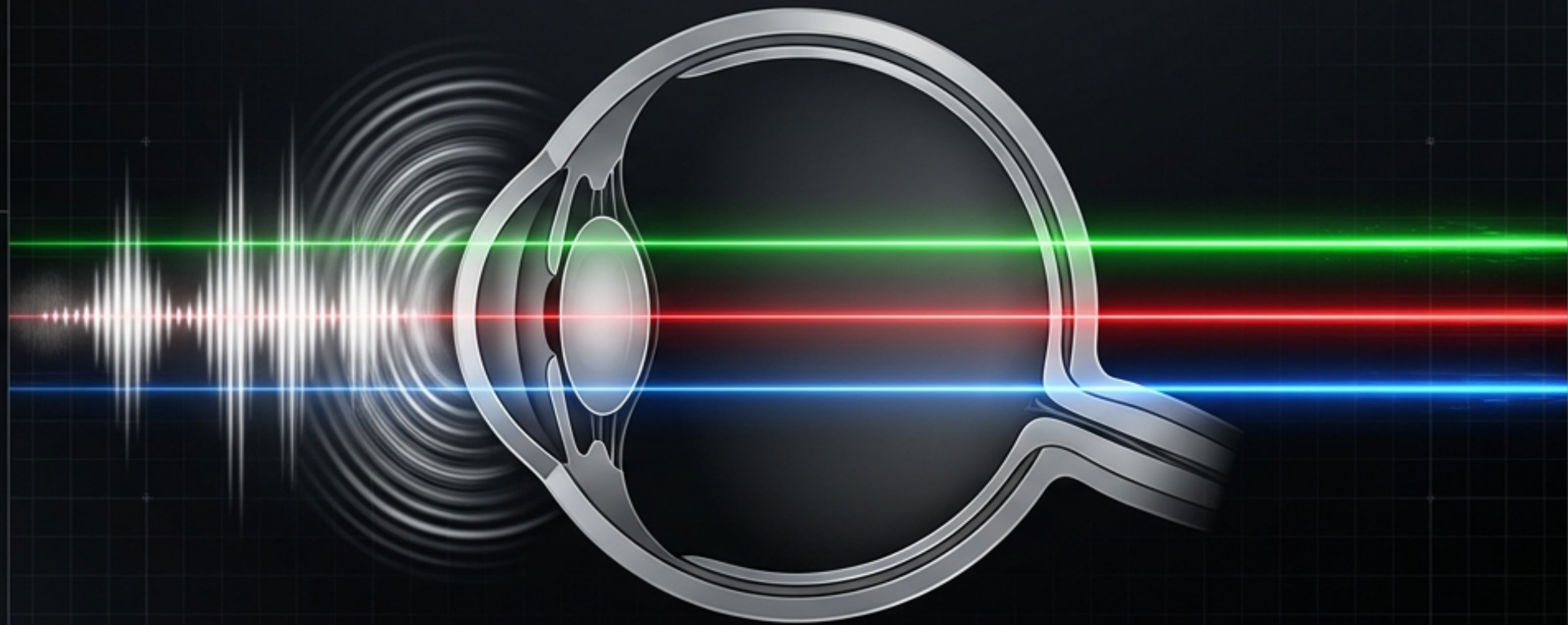


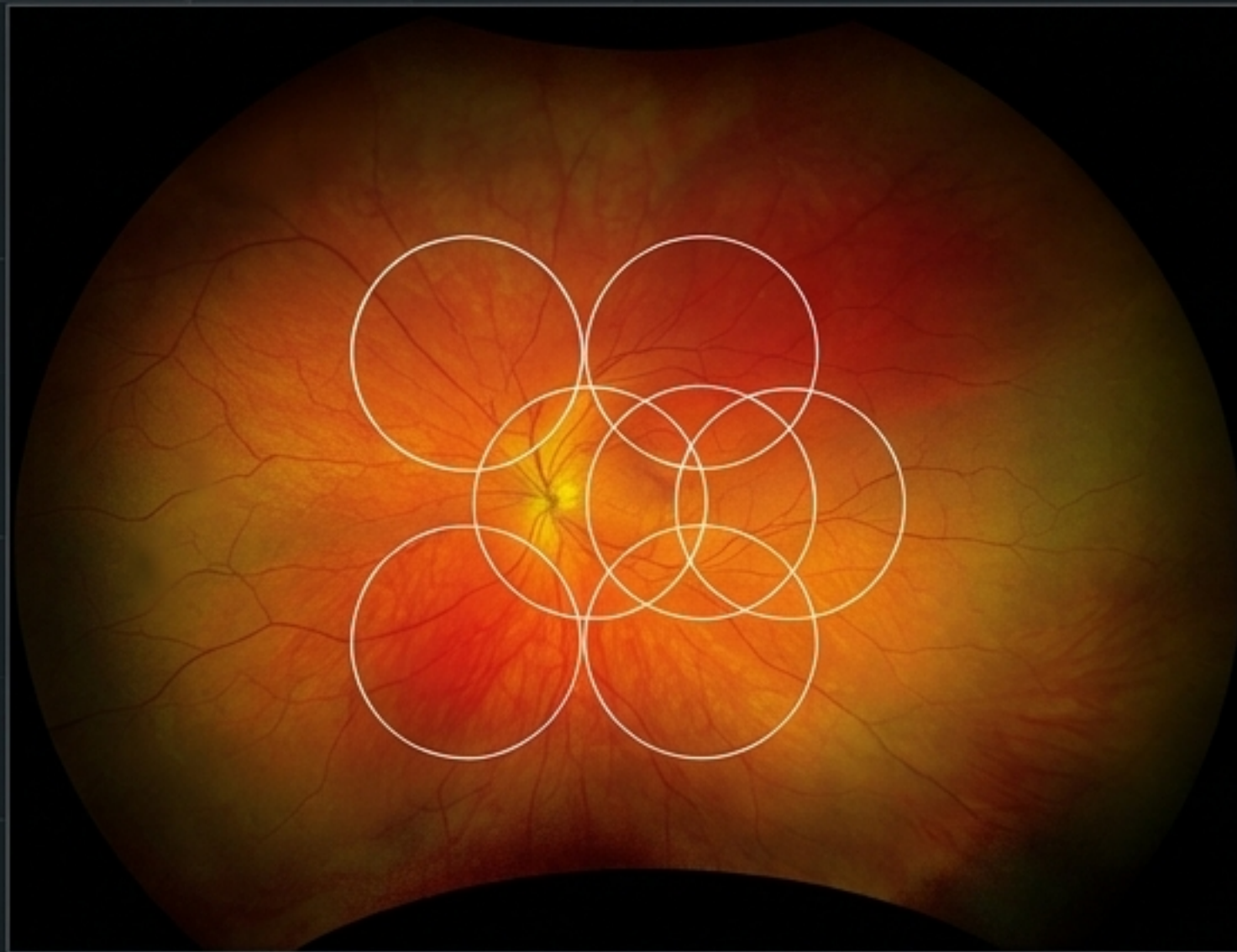
PIERCING THE VEIL

A CLINICAL MASTERCLASS IN ADVANCED OCULAR DIAGNOSTICS

Navigating Opaque Media, the Extreme Periphery, and Special Populations
through Acoustic and Photonic Imaging.

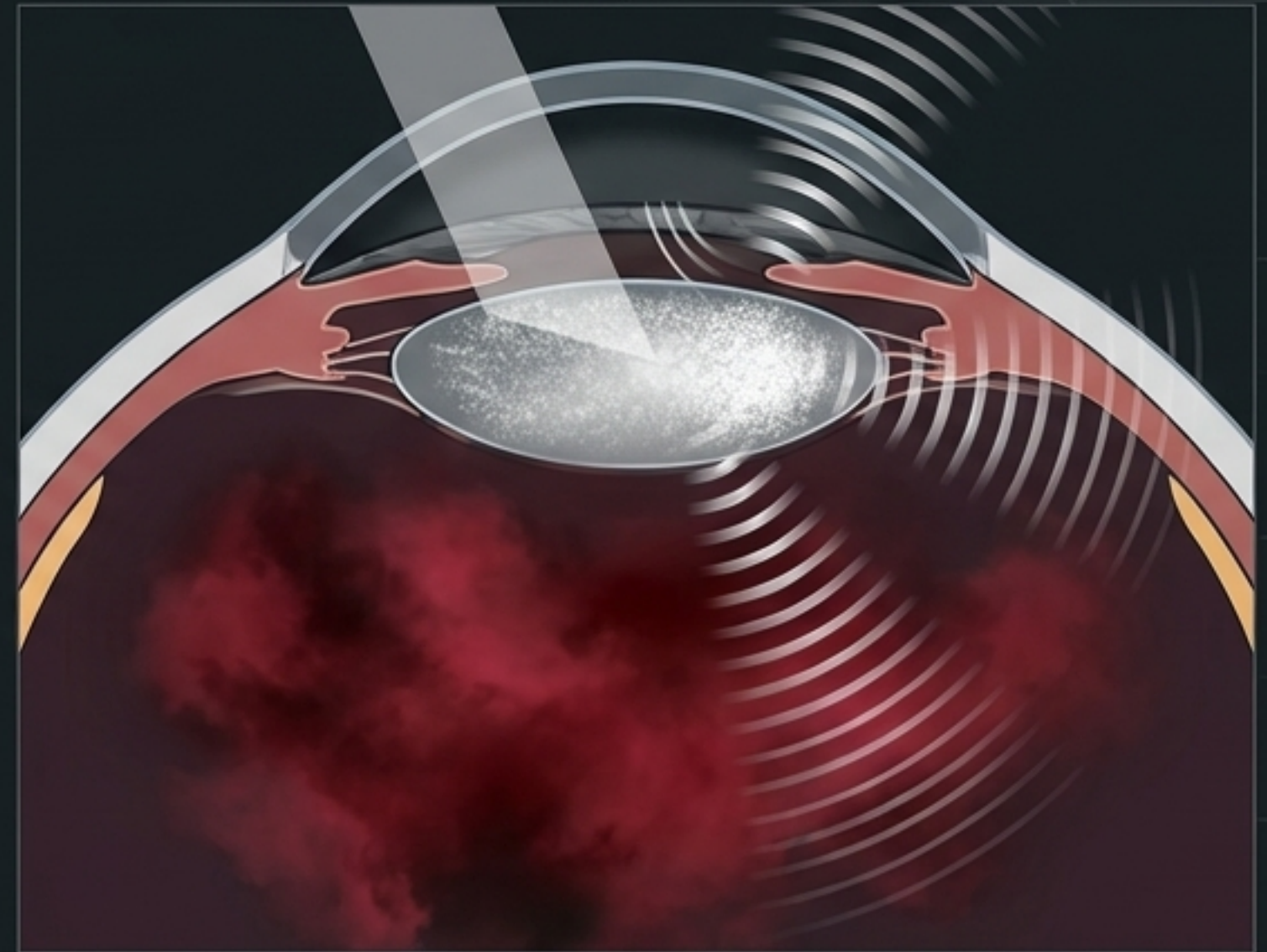


The Photonic Blind Spot



Standard 45° fundus photography only captures a fraction of the retina. The extreme periphery—where early diabetic lesions, tears, and progression indicators hide—remains in the dark.

The Acoustic Barricade



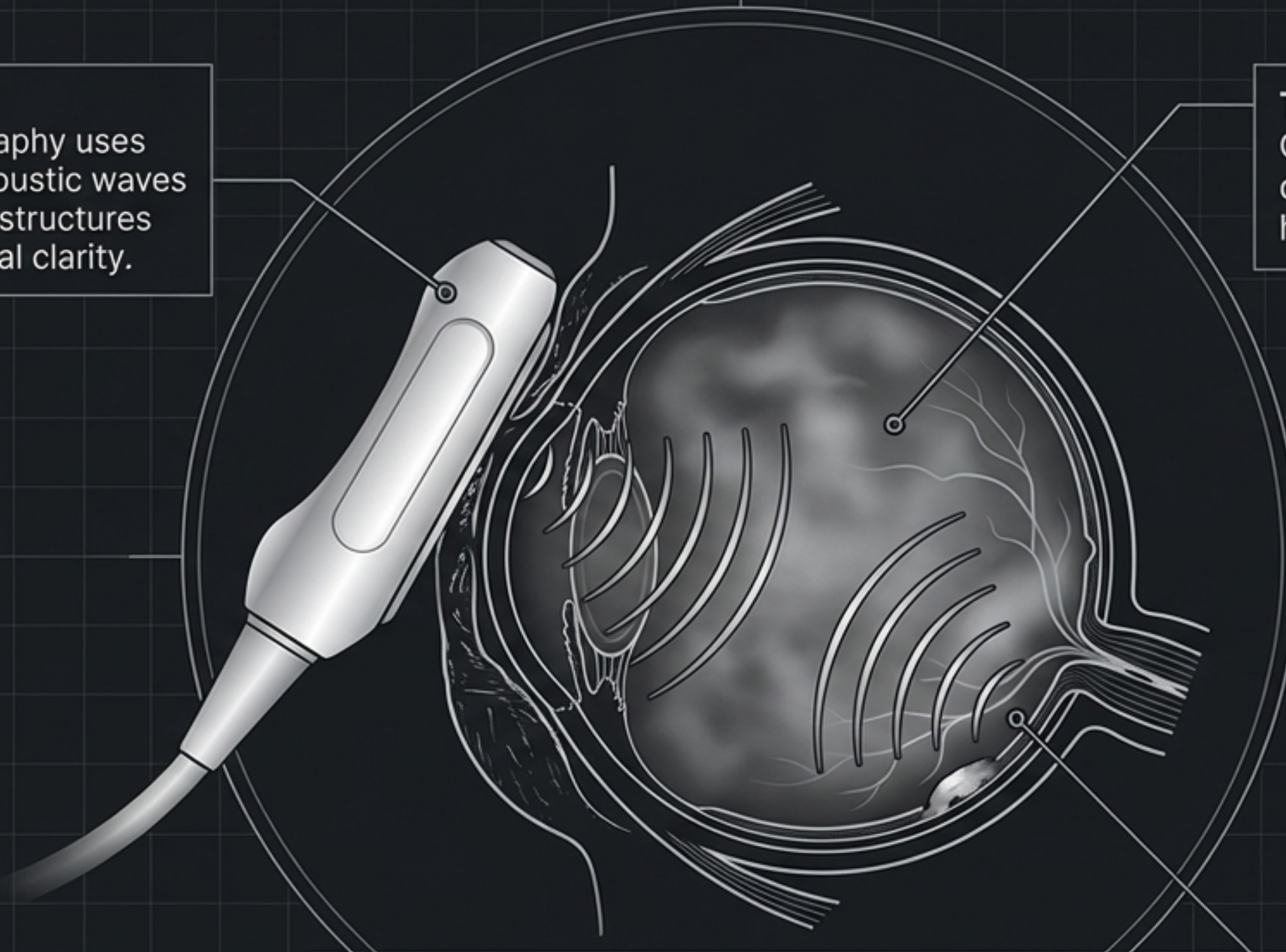
Standard Optical Coherence Tomography (OCT) relies on light transmission. Hazy media, uncooperative fixation, and small pupils cause severe signal attenuation, vignetting, and missing data.

Mechanism:

Gray-scale sonography uses high-frequency acoustic waves to map intraocular structures regardless of optical clarity.

The Advantage:

Completely bypasses cataracts, corneal edema, and vitreous hemorrhage.

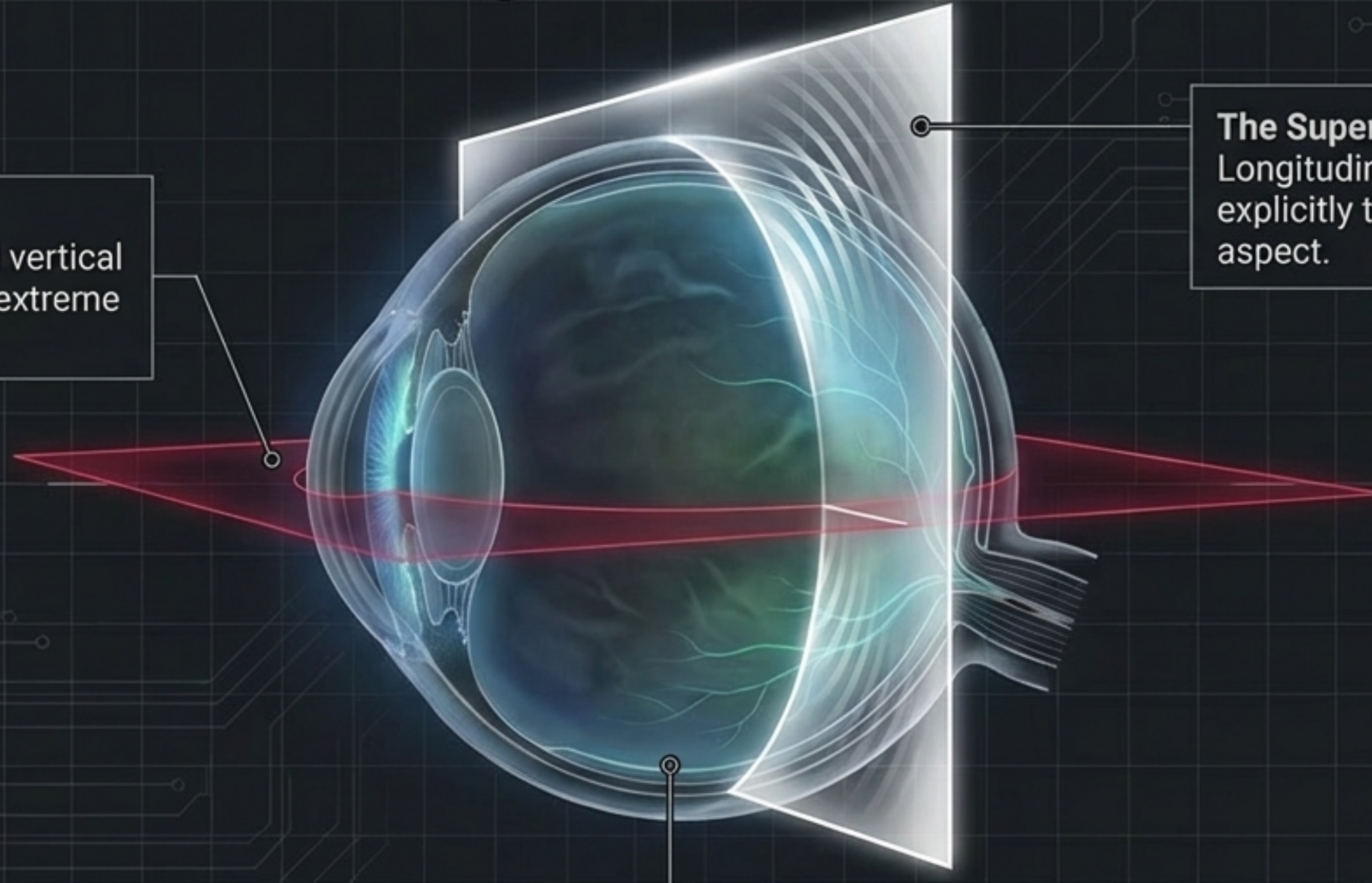
**Clinical Utility:**

Accurately identifies intraocular masses, retinal/choroidal detachments, and localizes foreign bodies even in emergently traumatized, closed eyes.

The Longitudinal Advantage

The Blind Spot:

Standard horizontal and vertical B-scans often miss the extreme anterior periphery.



The Superior Slice:

Longitudinal probe placement explicitly targets the superior aspect.

The Clinical Goal:

Detects subtle traction that causes retinal tears. Early detection prevents full retinal detachments. Also routinely detects macular edema and epi-macular membranes.

The B-Scan Pathology Matrix

	Retinal Detachment	PVD with Hemorrhage	Intraocular Mass/Tumor
SONOGRAPHIC APPEARANCE	Thick, highly reflective, continuous membrane.	Thin, discontinuous, or speckled presentation.	Solid, dome-shaped, or irregular lesion.
ECHOGENICITY	100% echogenicity (matches sclera).	Low to medium echogenicity.	Variable internal echogenicity; acoustic hollowing in melanomas.
MOVEMENT & KINETICS	Restricted, tethered mobility (often to the optic disc).	Highly mobile, fluid, 'swirling' after-movements.	Rigid, no independent movement.
KEY DIFFERENTIAL	-	Note: Longitudinal scans are critical to differentiate this from a true detachment.	-

The 200-Degree Reveal



Hidden Pathology

82%

Captures up to 82% of the retina in a single shot.

<0.5s

Ultra-fast capture time, significantly faster than traditional cameras.

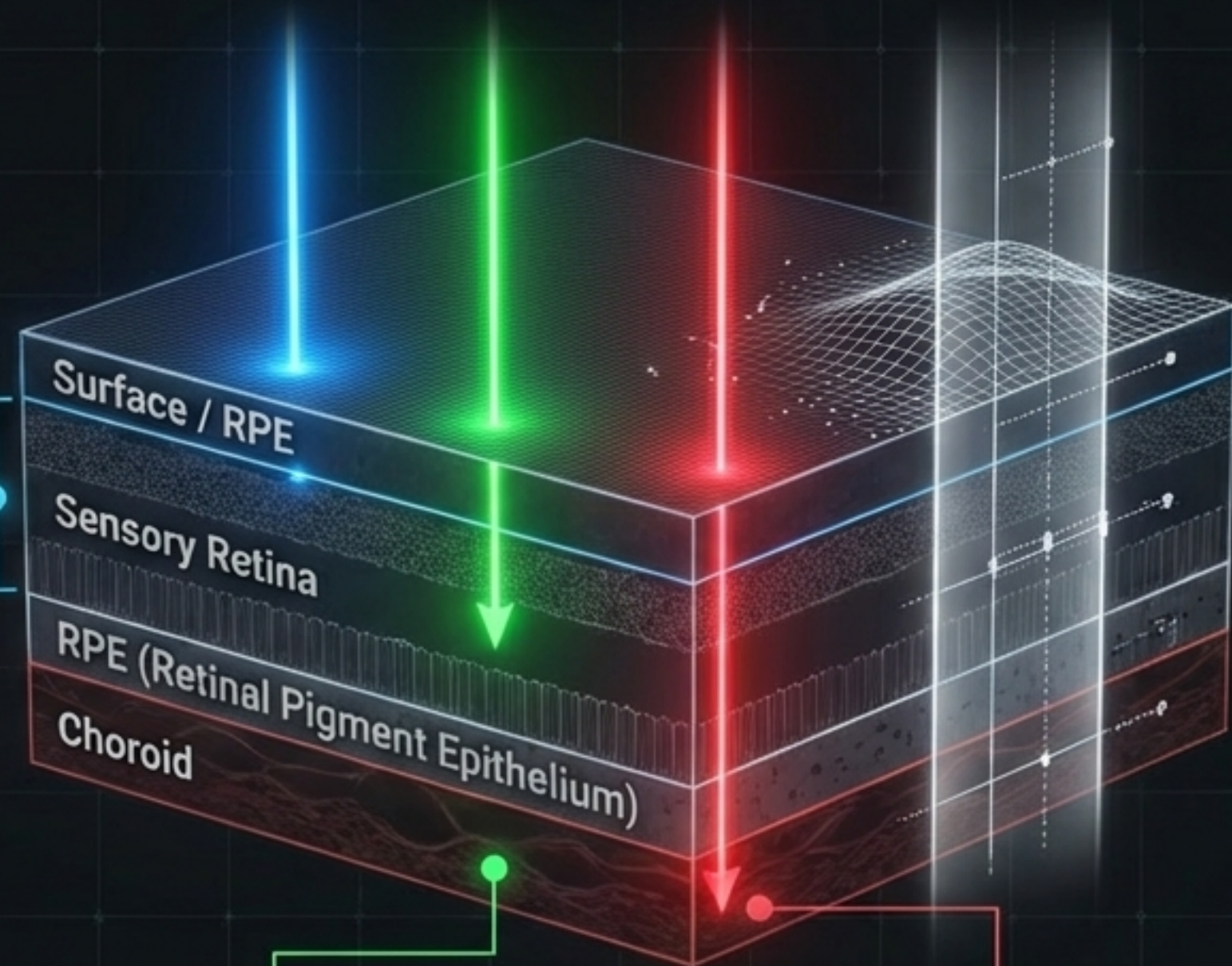
2 mm

No dilation required; images seamlessly through pupils as small as 2mm.

3+ Cataracts

Clinically proven to capture high-quality images through dense media opacities.

The Modality Cake Model



Neon Blue (Autofluorescence / FA):
Targets the surface and RPE for metabolic activity and vascular evaluation.

Neon Green (532 nm):
Penetrates to the Sensory Retina / RPE. Highlights nerve fibers and vascular structures.

Deep Red (633 nm):
Drives deep into the Choroid for deep tissue visualization.

1050 nm (Swept-Source OCT):
Provides deep, high-resolution cross-sectional topography anywhere in the retina.

The Silverstone Wavelength Matrix

Red Laser (633 nm)

Modality: optomap
Choroidal

Target: Deep tissue
visualization.

Combined: Teams
with Green and Blue
for optomap color rgb
(true-to-life color).

Green Laser (532 nm)

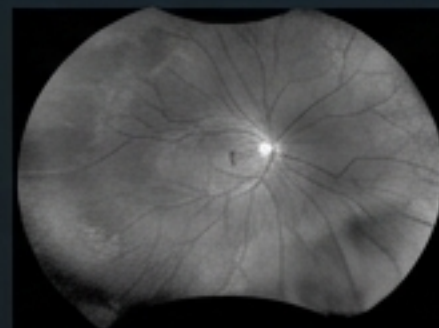
Modality: optomap
Sensory Retina &
Green AF

Target: Nerve fibers,
vascular structures,
RPE changes,
metabolic activity.

Blue Laser & Infra-Red

Modality: Blue AF,
Fluorescein
Angiography (FA),
ICG Angiography.

Target: Subtle retinal
pathology, vascular
evaluation, disease
detection.



Swept-Source OCT

Modality: Navigated
peripheral SS-OCT
(1050 nm).

Target: 4-in-1 color
depth imaging from
retinal surface via the
choroid.

Panoramic Autofluorescence (PAF)



The Mechanism

The **532 nm (Green)** and **Blue** lasers excite fluorophores in the retina, emitting a signal (570 to 780 nm) captured by the raster scan.

The Advantage

SLO optics are far less susceptible to hazy media than standard fundus cameras. Narrow optical pathways penetrate opacities.

Diagnostic Power

Instantly highlights retinal pigment epithelium (RPE) stress, structural loss, and pathophysiology in conditions like retinitis pigmentosa before they are visible in standard color photography.

The Pediatric & Special Populations Matrix

The Challenge	Traditional Method	The Optos UWF Protocol
The Drop-Phobic Child Extreme anxiety, refusal of dilating drops.	Incomplete exams, sedation, or forced compliance.	Non-mydriatic imaging captures 200° through a natural, undilated pupil.
Premature Infants (ROP) Fragile status, extremely small anatomical structures.	Scleral depression with indirect ophthalmoscopy.	Can be used in awake infants (with or without a speculum) to visualize the peripheral ridge of neovascularization and track regression.
Cognitive Impairment / Fixation Issues Inability to follow gaze instructions.	High rate of motion artifacts, ungradable images.	<0.5 second capture speed freezes motion, drastically reducing ungradable image rates.

Real-World Application Dashboard

Case A: 14-Month-Old with Leukocoria

History:

Left fundus normal, right eye presentation in the OR.

Intervention:

Fluorescein Angiography (FA) performed in video mode.

Diagnosis & Outcome:

Coats disease diagnosed. Full extent of retinovascular abnormalities identified and lasered. 7-month follow-up showed no leakage; retinal detachment prevented.

Case B: 10-Year-Old, Drop-Phobic

History:

Blurred vision, failed stem cell transplant. Refused dilating drops.

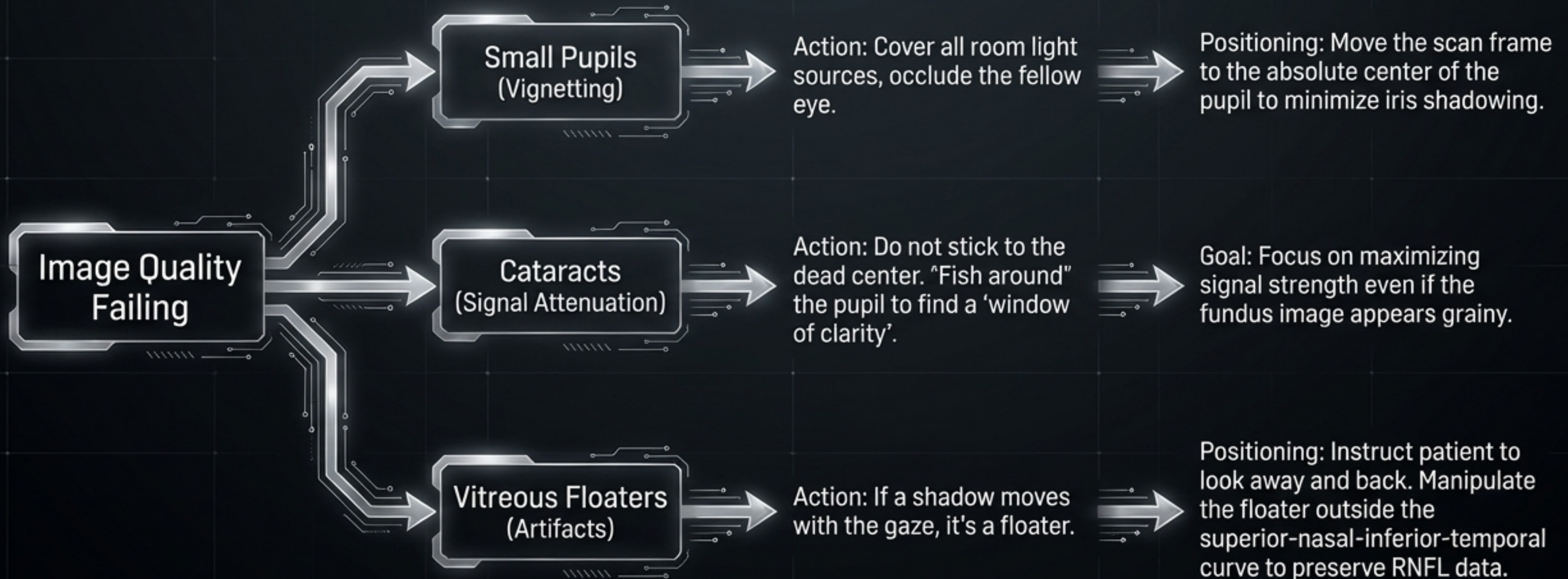
Intervention:

Optos imaging performed through an undilated pupil.

Diagnosis & Outcome:

Revealed bilateral disc swelling and a macular star. Diagnosed with idiopathic intracranial hypertension (IIH). Monitored successfully without drops.

Standard OCT Optimization Flowchart



Technician Tactics: Rescue Protocols

Managing the Surface

Dry Eye

Causes banding artifacts. Administer liquid artificial tears (avoid thick gels/ointments that disrupt signal). Instruct patient to close eyes between scans, blink once, then capture.

Fixation

For hyperactive children, use the 'Red Light, Green Light' method to lock gaze.

Software Tactics



Eye Tracking ON

Reduces blink artifacts and transient data loss. Use the 'track to prior scan' function for exact follow-up positioning.

Eye Tracking OFF

Use for faster acquisition in complex presentations (hazy media, extreme periphery, severe nystagmus). Requires a perfectly steady patient for a brief window.

The Master Diagnostic Matrix



The Opaque,
Closed, or
Traumatized Eye

Indicators: Dense cataracts, massive vitreous hemorrhage, emergency trauma.

B-Scan Ultrasound

Rationale: Acoustic waves ignore photonic barriers. Deploy longitudinal scans to rule out retinal detachments versus PVDs.



The Peripheral Threat
& Uncooperative Patient

Indicators: Diabetic screening, drop-phobic pediatrics, peripheral lesion monitoring.

Optos UWF (Silverstone)

Rationale: 200° capture in <0.5s through a 2mm pupil. Deploys 9 lasers/wavelengths to highlight metabolic stress and vascular issues.



The Macular & Central
Topography
Requirement

Indicators: Glaucoma tracking, central macular edema, epiretinal membranes.

Standard OCT / SS-OCT

Rationale: High-resolution cross-sectional layer analysis. Requires active technical optimization for opacities and dry eye.